Muhammad Tahir Zia

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Summary

Aspiring Digital IC Design & Verification Engineer with hands-on experience in SystemVerilog RTL design, RISC-V microarchitecture, and verification. Proficient in RTL design and simulation. Passionate about FPGA prototyping for complex digital systems.

Education & Training

Digital IC Design & Verification (INSPIRE) Bootcamp, Pakistan Software Export Board (PSEB) at GIKI — May 2026 – Present

- **Focus:** Intensive specialized training in RTL design, advanced processor microarchitecture, and functional verification methodologies.

BS in Computer Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan — Sept 2021 – June 2025

- **Coursework:** Fundamentals of Logic Design, Computer Organization and Assembly Language, Microprocessor Interfacing, Computer Communication and Networking, Signals & Systems, Digital Systems Design.

Technical Skills

Languages & Standards: SystemVerilog, Verilog, C/C++, RISC-V Assembly.

Verification & Design Concepts: SystemVerilog Layered Testbenches, FSM Design, Pipelining, Clock Domain Crossing (CDC), Static Timing Analysis.

EDA Tools & Platforms: Cadence Xcelium, Xilinx Vivado, EDA Playground, Nexys A7 FPGA.

Projects

32-bit RISC-V Pipelined Processor Core – *SystemVerilog, Vivado* (GitHub Repository)

- Architected an RV32I 5-stage pipelined processor, evolving from a foundational single-cycle datapath.
- Engineered a custom Hazard Unit to dynamically resolve data and control hazards via EX/MEM & MEM/WB forwarding, load-use stalling, and branch flushing.

SoC Hardware Accelerator – *SystemVerilog, Vivado* (GitHub Repository)

- Integrated a five-stage pipelined RISC-V processor core, alongside a specialized systolic array hardware accelerator tailored for image convolution operations with kernels.

Communication Protocols – *Verilog, Vivado, Nexys A7* (GitHub Repository)

- Implemented and deployed a UART controller on a NEXYS A7 FPGA while verifying I2C and SPI (Mode 0) protocols through Verilog FSM design and Vivado testbench simulations.

FPGA Morse Code Encoder – *Verilog, Vivado, Nexys A7* (GitHub Repository)

- Converts digits into its Morse code using an FSM based approach, displayed using LED lights. 1 second for dot and 3 seconds for Dash. Can be toggled into different modes for storing numbers in FIFO register as well.

Synchronous & Asynchronous FIFO – *SystemVerilog, Vivado* (GitHub Repository)

- Designed parameterized Synchronous and Asynchronous FIFOs in SystemVerilog, implementing Gray-coded pointers and 2-FF synchronizers for safe Clock Domain Crossing.

Final Year Project

BlockConnect (Social Media Application using Blockchain) — May 2024 – Jun 2025

- Developed a decentralized social networking layer by using Ethereum blockchain focused on data sovereignty and user privacy. Leveraged the architecture to eliminate the third-party data commoditization.